COMP1110 Project Plan

A Physics-Based Basketball Shot Simulation: Release Geometry, Backspin, and Ball Dynamics under Air Drag and Magnus Effects

1. Title and Team Members

Title: A Physics-Based Basketball Shot Simulation: Release Geometry, Backspin, and Ball Dynamics under Air Drag and Magnus Effects.

Team members (5): He Runlong (problem framing and literature), Li Zhenyi (projectile physics and integration), Shao Linjun (backboard / rim collision model), Liang Xuanshuo (dataset and validation), Chen Yin (visualisation and writing).

Course: COMP1110 Computing and Data Science in Everyday Life, 2026/27 Semester 1.

2. Problem Statement and Solution Concept

2.1 Motivation and Research Context

The prediction of basketball shooting outcomes has been studied by several researchers. Notably, Cao et al. (2017) demonstrated that OpenPose can estimate release parameters and predict shot outcomes, while Chen et al. (2020) used it to analyse shooting kinematics and provide posture correction feedback. These approaches focus on the shooters body mechanics rather than the physics of the ball in flight. Physics-based models (Brancazio, 1981; Alaways and Hubbard, 2002) study idealised trajectories but often neglect backspin effects or simplify rim/board collisions. Our project takes a different angle: we focus on the balls physics in flight - how release geometry, backspin, air drag, and Magnus lift jointly determine shot outcomes. This complements pose-estimation approaches by modelling what happens to the ball after release.

2.2 Problem Statement

A basketball in flight is a rigid-body problem: its trajectory is set by three release parameters - release point, release force (initial velocity vector), and release backspin - and shaped by gravity, air drag, and Magnus lift from spin. When the ball meets the backboard or rim, the outcome depends on geometric collision with rigid surfaces, with kinetic energy lost at each bounce.

Research question: Given a rigid, immovable backboard and rim with a fixed restitution constant, can a pure-physics simulation driven by release point, release force vector, and

release backspin (with air drag and Magnus lift) reproduce whether a shot (a) scores a swish, (b) bounces off the backboard and goes in, or (c) misses? How large is the models disagreement with real-world measured shots?

2.3 Scope

In scope:

- Three release parameters: release point (x, y, z), release force vector (v0x, v0y, v0z), initial backspin rate (omega).
- Air drag (quadratic) and Magnus lift from backspin.
- Rigid, immovable backboard (vertical plane) and rigid, immovable rim (torus); inelastic collisions with a single restitution constant e = 0.7.
- Scoring rule: swish, rim-and-in, board-and-in, and miss outcomes.
- Extension: comparison of free-throw vs. 3-point shot dynamics, examining how release distance and launch angle alter the importance of air drag and Magnus effects.

Out of scope:

- Ball deformation, rim flexing, backboard vibration.
- Player physiology, fatigue, or psychological factors.
- Side-spin and wobble; we focus on backspin only.
- Machine-learning correction of physics equations - the model is pure physics.
- Real-time pose estimation or video analysis.

3. Computing and Data Science Connection

Main project type: algorithmic / simulation-modelling (pure physics), with a small data-analysis component for model validation.

Key concepts and tools:

- Numerical integration (Python, numpy, RK4 or semi-implicit Euler) - core of the simulation.
- Air drag and Magnus lift equations - convert release parameters into continuous trajectories.
- Rigid-body collision detection - plane vs sphere (backboard), torus vs sphere (rim), with restitution constant e.
- Outcome labelling - swish, rim-bounce-in, board-bounce-in, and miss classification.
- Parameter sweep - grid search over release angle and backspin, producing a make-probability surface.
- External dataset (NBA SportVU / Kaggle / own 20-30 filmed free throws) - used only for validation, not training.
- Error quantification - reporting disagreement rate between predicted and recorded outcomes.

Technical approaches: We adopt a calculation-based (physics simulation) approach as the core method, estimating physical parameters like air resistance and ball spin from video footage using computer vision. An AI-based (data-driven) approach - requiring many multi-angle shooting videos and ML models - is discussed as a potential extension but deemed beyond feasible scope for this course project.

The project is not machine-learning based. Every prediction is the result of solving physics equations with fixed constants.

4. Literature / Background Review

We will cover roughly 10-15 sources, organised into three thematic clusters:

Cluster 1: Physics of basketball and ball dynamics.

- Brancazio, P. J. (1981). Physics of basketball. Am. J. Phys., 49(4), 356-365. - Classical projectile model with fixed restitution.
- Alaways, L. W., & Hubbard, M. (2002). Flight dynamics of the basketball. Sports Eng., 5(1), 29-37. - Drag and Magnus lift coefficients for a spinning basketball.
- Nürnberg, S., et al. (2021). A validated simulation model of basketball free-throw trajectories. Sports Eng., 24, 1-12. - Template for error quantification.
- Hunt, J. C. (2011). Projectile dynamics in sport. Human Kinetics. - Reference for restitution and inelastic contact modelling.

Cluster 2: Computer vision and pose estimation.

- Cao, Z., et al. (2017). OpenPose: Realtime multi-person 2D pose estimation. arXiv:1611.08050. - Pose-estimation pipeline for works like Chen et al. (2020).
- Chen, Y., et al. (2020). Prediction of basketball free throw shooting by OpenPose. - Vision-based release parameter estimation; distinct from our ball-physics approach.

Cluster 3: Kinematics and data sources.

- Okazaki, V. H. A., et al. (2011). A review on basketball shooting kinematics. Rev. Bras. Educ. Fís. Esporte, 25(4), 665-680. - Empirical release parameter ranges.
- NBA SportVU / AWS public tracking datasets. - External validation data.
- Our own filmed dataset (20-30 free throws, mobile camera at 60 fps).
- Wiley, S., et al. (2014). Release conditions for successful and unsuccessful free throws. J. Sports Sci., 32(12), 1148-1154. - Calibrating parameter sweep ranges.

Research gap: Most existing simulations either (a) ignore backspin and Magnus lift, (b) do not separate swish from rim-bounce outcomes, or (c) treat the rim as flexible. The OpenPose approach focuses on posture correction rather than ball physics. No recent study combines (i) a single fixed restitution constant, (ii) rigid board and rim, (iii) explicit swish vs. rim-bounce vs. board-bounce comparison, and (iv) a single error number against external data. This benchmark is our contribution.

5. Feasibility, Limitations, and Ethics

Feasibility: The equations of motion (drag + Magnus + gravity) are well-established and implementable in Python/numpy quickly. RK4 integration and sphere-plane/torus collision detection are standard problems with reference implementations. A single smartphone at 60 fps suffices for data collection. Parameter sweeps over a grid of release conditions complete in minutes to hours on a standard laptop.

Constants and data: All physics constants (g, rho_air, C_d, S, ball mass, ball radius, restitution e, board/rim geometry) are hard-coded as Python consts and frozen after initial derivation - not tuned to the dataset. Validation uses NBA SportVU open data and 20-30 filmed free throws.

Limitations we will explicitly state:

- Rigid board and rim - real rims flex and boards vibrate; rim-bounce-in outcomes will be less accurate than swish outcomes.
- Single restitution constant e is a simplification; real board and rim contacts have different e values.
- Spin decay in air beyond Magnus force is not modelled; spin is held constant during flight.
- Small empirical sample - the error number is a sanity check, not a statistically strong claim.
- We will not claim the model predicts real-player performance - only that it is physically consistent under stated simplifications.

Ethics: All filmed participants are group members who give verbal consent. Footage is used only to label shot outcomes; no identifying face data is required. Footage is stored locally and deleted after the final report. NBA SportVU data is public and anonymised.

6. Timeline and Team Roles

Phase 1 (Weeks 1-3): Literature review and physics model design. He Runlong and Li Zhenyi derive equations of motion and set up numerical integration. Shao Linjun researches collision detection. Liang Xuanshuo sources SportVU data and sets up filming. Chen Yin begins introduction and literature review.

Phase 2 (Weeks 4-6): Core simulation. Li Zhenyi implements RK4 integrator with drag and Magnus effects. Shao Linjun implements backboard and rim collision detection. He Runlong validates against Brancazio (1981). Liang Xuanshuo begins filming. Chen Yin drafts methods section.

Phase 3 (Weeks 7-9): Extension and validation. Li Zhenyi extends simulation to 3-point shots. Liang Xuanshuo processes footage and extracts outcome labels. All members contribute to error analysis and validation.

Phase 4 (Weeks 10-12): Integration, visualisation, and writing. Chen Yin coordinates final report and figure consistency. All members review and refine. Final submission and presentation.

Roles may shift as the project progresses, but the named owners are the first point of contact for each milestone.

7. Expected Outcomes

- Working Python simulation predicting free-throw outcomes from release geometry and backspin.
- Make-probability heatmap showing success variation with release angle, height, and backspin.
- Quantitative validation against real-world dataset, reporting agreement rate.
- Comparative analysis of free-throw vs. 3-point shot physics.
- Preliminary feasibility assessment of an AI-based extension for future work.